

MINI ASSESSMENT

Name: Kurikuti Harathi

Reg No: 99230040981

Department: CSE AIML

1. If an API returns 429, what should your program do?

Answer: A 429 status code means that the API has received too many requests in a short period and is temporarily limiting further requests. The program should avoid retrying immediately because that can increase the problem. It should wait for some time and then try the request again. If the server sends a **Retry-After** value, the program should respect that waiting time. For repeated 429 responses, an increasing delay between retries can be used. This helps reduce unnecessary traffic and gives the API enough time to recover.

2. If `response.json()` fails, what should your program do?

Answer: If `response.json()` fails, it means the response could not be converted into valid JSON. The server might have returned plain text, HTML, an empty response, or another unexpected format. The program should handle this situation instead of terminating unexpectedly. A **try/except** block can be used to catch the JSON parsing error and provide a suitable fallback or error message. It is also a good practice to check the HTTP response status before attempting to read the JSON data.

3. If a JSON key is missing, why might `.get()` be safer than `[]`?

Answer: The `[]` operator directly expects the specified key to exist in the dictionary. For example, `data['name']` raises a **KeyError** when the key is missing, which can interrupt the program. In contrast, `data.get('name')` returns **None** when the key is not available, allowing the program to continue safely. A default value can also be supplied, such as `data.get('name', 'Unknown')`. Therefore, `.get()` is useful when working with API or JSON data where some fields may be optional or absent.